

Solutions

1. **Solution:** Readings cluster tightly $\Rightarrow$ high **precision**. Mean $\approx 20.0^\circ\text{C}$ vs. true $19.9^\circ\text{C} \Rightarrow$ fairly **accurate** (small bias).
2. **Solution:** Abs. error $|49.6 - 50.0| = 0.4$ cm. Percent error $= 0.4/50.0 = 0.008 = 0.8\%$.
3. **Solution:** Nearest 0.01 m $\Rightarrow$ interval $[7.84 - 0.005, 7.84 + 0.005) = [7.835, 7.845)$ m.
4. **Solution:** Nearest 0.1 cm $\Rightarrow [3.35, 3.45)$ cm.
5. **Solution:** 1.25 (nearest 0.01) $\Rightarrow [1.245, 1.255)$; 2.4 (nearest 0.1) $\Rightarrow [2.35, 2.45)$. Sum interval $= [1.245 + 2.35, 1.255 + 2.45) = [3.595, 3.705)$ m.
6. **Solution:** $L \in [4.195, 4.205)$, $W \in [1.75, 1.85)$. Area interval $\in [4.195 \cdot 1.75, 4.205 \cdot 1.85) \approx [7.341, 7.779)$ m² (endpoints via min/max product since both positive).
7. **Solution:** (a) 3 sig figs; (b) ambiguous (usually 2, unless written 1.200×10^3); (c) 4 sig figs (coefficient 1.200 has 4); (d) 3 sig figs (decimal point makes trailing zeros significant).
8. **Solution:** Exact sum $= 12.07 + 0.3 + 0.025 = 12.395$. Least precise place is 0.1 (from 0.3), so 12.4.

9. **Solution:** Exact product = $3.20 \times 0.054 = 0.1728$. Sig figs: $\min(3, 2) = 2 \Rightarrow 0.17 = 1.7 \times 10^{-1}$.
10. **Solution:** First $7.16 - 3.2 = 3.96 \rightarrow$ round to 0.1 place (3.2 limits) $\Rightarrow 4.0$. Then $4.0/0.041 \approx 97.56$. Sig figs for division: $\min(2, 2) = 2 \Rightarrow 98$.
11. **Solution:** Balance B (to 0.01 g) gives higher precision (smaller spread). Relative step $\approx 0.01/25 = 0.0004 = 0.04\%$ vs $0.1/25 = 0.4\%$.
12. **Solution:** Very **precise** (tight clustering) but **inaccurate** (far from 10.50 cm).
13. **Solution:** $r \in [12.35, 12.45)$ cm. Then $C \in [2\pi(12.35), 2\pi(12.45)) \approx [77.6, 78.2)$ cm.
14. **Solution:** $s \in [3.55, 3.65)$ cm $\Rightarrow A \in [3.55^2, 3.65^2) = [12.60, 13.32)$ cm².
15. **Solution:** Nearest meter $\Rightarrow$ abs. error ≤ 0.5 m. Worst case percent error $\frac{0.5}{142} \approx 0.352\%$.
16. **Solution:** (a) 4.57×10^{-4} (b) 6.02×10^6 (c) 3.00×10^2 .
17. **Solution:** Unrounded product: $9.4 \times 2.15 = 20.21$. Sig-figs: $\min(2, 3) = 2 \Rightarrow 20$ (two sig figs). Range: $9.4 \rightarrow [9.35, 9.45)$, $2.15 \rightarrow [2.145, 2.155)$, so $A \in [9.35 \cdot 2.145, 9.45 \cdot 2.155) \approx [20.05, 20.36)$ cm².

18. **Solution:** Round sum to least precise place: 0.01 place (from 0.27). Exact: $3.418 + 0.27 = 3.688 \Rightarrow 3.69$ m.

19. **Solution:** $A = 100.0 \pm 0.05$ cm (step 0.1) $\Rightarrow$ rel. $\approx 0.05/100 = 0.0005 = 0.05\%$.
 $B = 10.00 \pm 0.005$ cm $\Rightarrow$ rel. $\approx 0.005/10 = 0.0005 = 0.05\%$. Here they are equal by design of last digit; if step sizes matched absolute decimal places, shorter values often have larger relative error.

20. **Solution:** Sig-fig: $m = 12.0$ (3 s.f.), $V = 4.0$ (2 s.f.) $\Rightarrow \rho = 12.0/4.0 = 3.0$ g/cm³ (2 s.f.). Interval: $m \in [11.95, 12.05)$, $V \in [3.95, 4.05)$, so $\rho \in [\frac{11.95}{4.05}, \frac{12.05}{3.95}) \approx [2.95, 3.05)$ g/cm³.

21. **Solution:** $r \in [7.95, 8.05)$; $A = \pi r^2$ increases with r , so use endpoints: $A \in [\pi(7.95^2), \pi(8.05^2)) \approx [198.5\pi, 204.0\pi)$ cm².

22. **Solution:** Display 2.40 V implies nearest 0.01 V $\Rightarrow$ interval $[2.395, 2.405)$ V; sig figs = 3.

23. **Solution:** Numerator: $1.205 + 0.88 = 2.085 \rightarrow$ add/sub rounds to 0.01 place (0.88 limits) $\Rightarrow 2.09$ (3 s.f.). Then $2.09/0.0315 \approx 66.349\dots$; sig figs $\min(3, 3) = 3 \Rightarrow 66.3 = 6.63 \times 10^1$.

24. **Solution:** Best practice: keep guard digits; round once at the end. Example: $(4.37 + 1.2) \times 0.015$. Exact sum = 5.57 $\rightarrow$ round to 0.1 place $\Rightarrow 5.6$; $5.6 \times 0.015 = 0.084$ (2 s.f.). If you had prematurely rounded $4.37 \rightarrow 4.4$, sum = 5.6 anyway here, but in general premature rounding can shift the last digit; keep extra digits until the final step.

25. **Solution:** (a) $x \in [5.195, 5.205)$, $y \in [1.25, 1.35)$, $z \in [0.835, 0.845)$.
(b) $V \in [5.195 \cdot 1.25 \cdot 0.835, 5.205 \cdot 1.35 \cdot 0.845) \approx [5.422, 5.936) \text{ cm}^3$.
(c) Sig-figs for product: 5.20 (3 s.f.), 1.3 (2 s.f.), 0.84 (2 s.f.) $\Rightarrow$ 2 s.f. Report $V = (5.20)(1.3)(0.84) = 5.6832 \rightarrow \boxed{5.7 \text{ cm}^3}$. Interval report: $\boxed{5.422 \leq V < 5.936 \text{ cm}^3}$.